

Rovibrational analysis: Li_Omega

Sinc-DVR (Colbert–Miller) + sixth-degree Rydberg potential

Reduced mass $\mu = 12549.06992 m_e$ (6.88417 amu). Grid: 500 sinc-DVR points on the CSV range. Level count is automatic: all bound states below D_e .

Table 1: Optimized sixth-degree Rydberg parameters (atomic units unless noted).

Parameter	Value
R_e (bohr)	3.314473
R_e (Å)	1.753944
D_e (hartree)	0.08139736
D_e (cm ⁻¹)	17864.656
D_e (kcal mol ⁻¹)	51.0776
$V(R_e)$ (hartree)	-0.08385014
c_1	0.64445265
c_2	-0.04457551
c_3	-0.00346911
c_4	0.00837446
c_5	-0.00100194
c_6	0.00006553
fit RMS (hartree)	8.493e-06

Table 2: Vibrational energy levels, spectrum $E_v - E_0$ and level differences ΔE_v (cm⁻¹).

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
0	197.7921	199.3792	0.0000	392.3450
1	590.1371	591.7143	392.3450	387.6556
2	977.7927	979.3598	780.0006	382.9413
3	1360.7340	1362.2907	1162.9419	378.2020
4	1738.9360	1740.4822	1541.1439	373.4376
5	2112.3736	2113.9089	1914.5815	368.6478
6	2481.0214	2482.5455	2283.2293	363.8324
7	2844.8538	2846.3666	2647.0617	358.9915
8	3203.8453	3205.3464	3006.0532	354.1248
9	3557.9701	3559.4594	3360.1780	349.2324
10	3907.2025	3908.6796	3709.4104	344.3142
11	4251.5167	4252.9814	4053.7246	339.3703
12	4590.8870	4592.3390	4393.0949	334.4008
13	4925.2878	4926.7269	4727.4957	329.4058
14	5254.6935	5256.1194	5056.9015	324.3855
15	5579.0790	5580.4914	5381.2869	319.3402
16	5898.4192	5899.8177	5700.6271	314.2703
17	6212.6895	6214.0739	6014.8974	309.1762
18	6521.8656	6523.2357	6324.0735	304.0584

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
19	6825.9241	6827.2795	6628.1320	298.9178
20	7124.8418	7126.1823	6927.0497	293.7549
21	7418.5967	7419.9220	7220.8046	288.5707
22	7707.1674	7708.4772	7509.3754	283.3664
23	7990.5338	7991.8278	7792.7418	278.1431
24	8268.6769	8269.9548	8070.8848	272.9022
25	8541.5791	8542.8406	8343.7870	267.6453
26	8809.2244	8810.4693	8611.4323	262.3742
27	9071.5986	9072.8266	8873.8065	257.0910
28	9328.6896	9329.9004	9130.8975	251.7979
29	9580.4874	9581.6808	9382.6954	246.4975
30	9826.9849	9828.1606	9629.1928	241.1925
31	10068.1774	10069.3351	9870.3853	235.8861
32	10304.0635	10305.2031	10106.2715	230.5818
33	10534.6454	10535.7665	10336.8533	225.2833
34	10759.9286	10761.0312	10562.1366	219.9946
35	10979.9232	10981.0070	10782.1312	214.7202
36	11194.6435	11195.7082	10996.8514	209.4650
37	11404.1084	11405.1540	11206.3163	204.2339
38	11608.3423	11609.3686	11410.5502	199.0324
39	11807.3747	11808.3816	11609.5826	193.8664
40	12001.2411	12002.2285	11803.4490	188.7417
41	12189.9828	12190.9506	11992.1907	183.6647
42	12373.6475	12374.5958	12175.8554	178.6418
43	12552.2893	12553.2180	12354.4973	173.6796
44	12725.9689	12726.8779	12528.1768	168.7845
45	12894.7534	12895.6429	12696.9613	163.9632
46	13058.7166	13059.5866	12860.9245	159.2219
47	13217.9385	13218.7891	13020.1464	154.5669
48	13372.5054	13373.3368	13174.7133	150.0038
49	13522.5092	13523.3216	13324.7171	145.5380
50	13668.0472	13668.8407	13470.2551	141.1741
51	13809.2213	13809.9963	13611.4292	136.9164
52	13946.1377	13946.8943	13748.3456	132.7681
53	14078.9057	14079.6443	13881.1137	128.7318
54	14207.6375	14208.3584	14009.8454	124.8094
55	14332.4469	14333.1503	14134.6548	121.0019
56	14453.4487	14454.1351	14255.6567	117.3095
57	14570.7582	14571.4278	14372.9661	113.7317
58	14684.4899	14685.1431	14486.6978	110.2674
59	14794.7573	14795.3945	14596.9652	106.9149
60	14901.6722	14902.2937	14703.8801	103.6718
61	15005.3440	15005.9502	14807.5519	100.5355
62	15105.8795	15106.4708	14908.0874	97.5031

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
63	15203.3826	15203.9594	15005.5905	94.5715
64	15297.9541	15298.5166	15100.1620	91.7372
65	15389.6913	15390.2399	15191.8992	88.9970
66	15478.6883	15479.2234	15280.8962	86.3476
67	15565.0359	15565.5578	15367.2439	83.7859
68	15648.8218	15649.3308	15451.0297	81.3086
69	15730.1304	15730.6269	15532.3384	78.9130
70	15809.0434	15809.5277	15611.2514	76.5963
71	15885.6397	15886.1120	15687.8476	74.3559
72	15959.9956	15960.4563	15762.2035	72.1894
73	16032.1850	16032.6344	15834.3929	70.0948
74	16102.2798	16102.7181	15904.4877	68.0698
75	16170.3495	16170.7771	15972.5574	66.1125
76	16236.4620	16236.8792	16038.6700	64.2211
77	16300.6831	16301.0901	16102.8911	62.3937
78	16363.0768	16363.4739	16165.2847	60.6284
79	16423.7051	16424.0926	16225.9131	58.9233
80	16482.6285	16483.0066	16284.8364	57.2766
81	16539.9051	16540.2742	16342.1130	55.6863
82	16595.5914	16595.9516	16397.7993	54.1502
83	16649.7416	16650.0932	16451.9495	52.6660
84	16702.4076	16702.7509	16504.6155	51.2316
85	16753.6392	16753.9744	16555.8471	49.8444
86	16803.4836	16803.8108	16605.6915	48.5018
87	16851.9854	16852.3050	16654.1933	47.2014
88	16899.1868	16899.4989	16701.3947	45.9403
89	16945.1271	16945.4318	16747.3350	44.7164
90	16989.8435	16990.1411	16792.0514	43.5321
91	17033.3756	17033.6664	16835.5835	42.4207
92	17075.7963	17076.0808	16878.0042	41.5438
93	17117.3401	17117.6203	16919.5480	41.3234
94	17158.6635	17158.9444	16960.8714	—

Table 3: Spectroscopic constants (cm^{-1}), finite differences of the first levels (Eq. 7 of Silva *et al.*, J. Mol. Model. **24**, 235, 2018).

Constant	J=0	J=1
ω_e	397.0107	397.0011
$\omega_e x_e$	2.32613	2.32625
$\omega_e y_e$	-4.1330e-03	-4.1336e-03
B_e	0.795987	
α_e	4.8271e-03	
γ_e	-5.9032e-05	

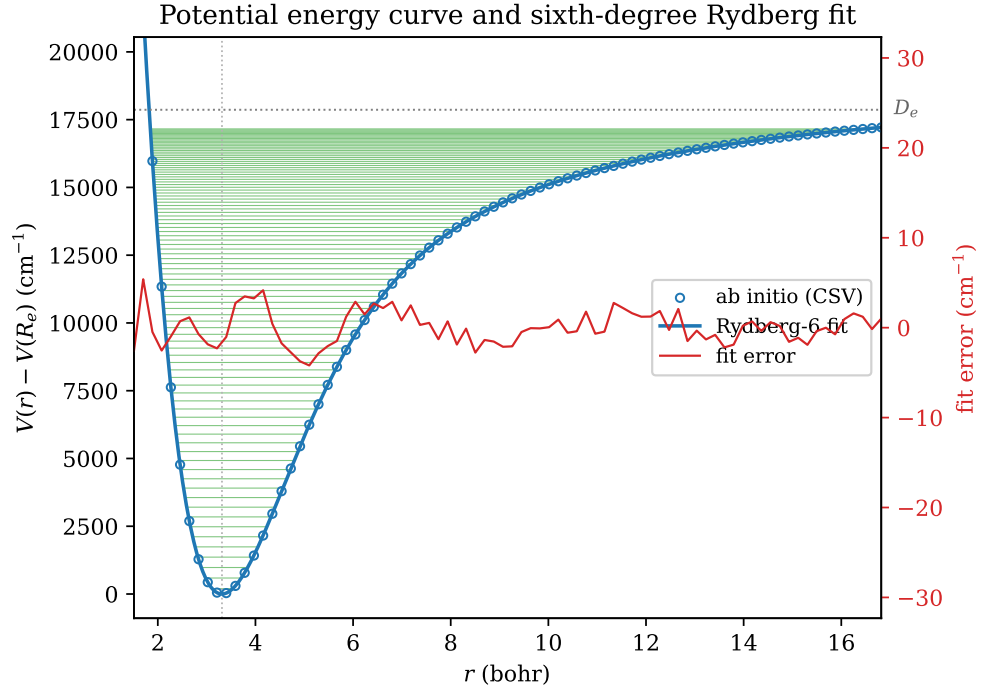


Figure 1: Potential energy curve: ab initio points, sixth-degree Rydberg fit, fit error (right axis, red), and the $J=0$ vibrational levels (green).

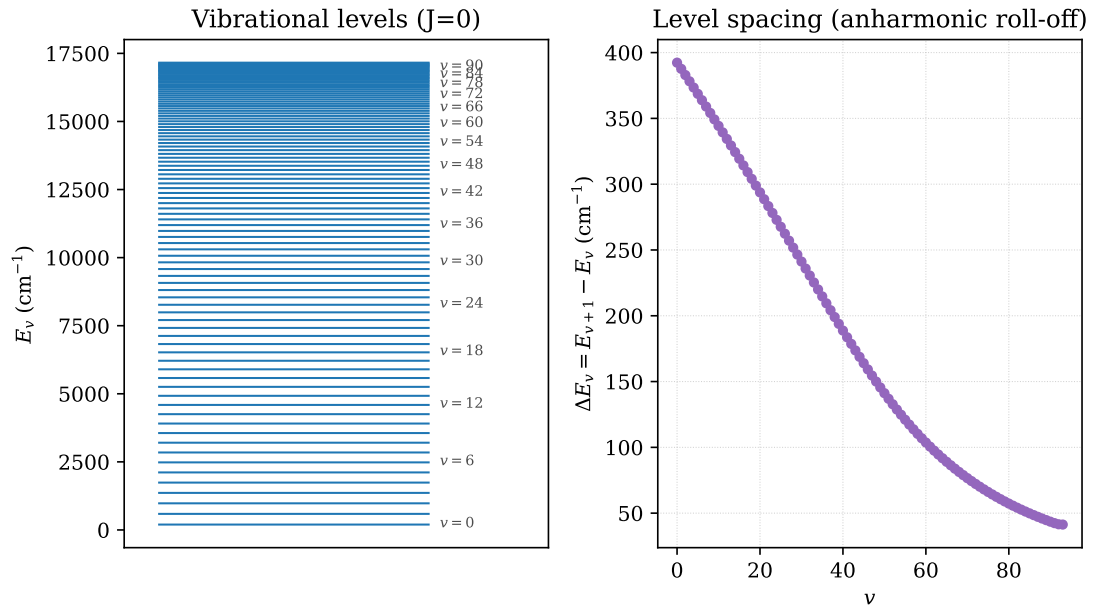


Figure 2: Vibrational spectrum: energy-level ladder (left) and the anharmonic decrease of the level spacing ΔE_v (right).